Graham Anderson

Buffalo, NY · Buffalo-area hybrid/onsite or remote
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I build agent systems that can prove what they did.

Principal AI Engineer | Machine Learning Engineer | AI Architect | Agentic AI · LLM/RAG · Knowledge Graphs | Defense & Aerospace | Founder, grahamaco | Buffalo, NY · Remote

U.S. citizen. Extensive experience delivering under export-controlled (ITAR) constraints.

ABOUT

An unusual résumé: commercial composer for Adidas and Pepsi, Webby-recognized producer for Sony, DARPA technical lead alongside Lockheed Martin and MIT. High-end creative and hard technical work — delivered by the same person, shipped as working code, in public.

I build verifiable agentic AI systems for defense, aerospace, and compliance. On DARPA ARCOS, I was Principal Data Scientist and ACERT Technical Lead for CS Group's contribution: the responsible technical/project lead for knowledge-graph and LLM reasoning delivery for automated certification of mission-critical software, working alongside Honeywell, Lockheed Martin, MIT, GE Research, SRI, and other program collaborators.

Much of my current client work is export-controlled, so client and program details are withheld. Publicly releasable independent engineering is at github.com/grahama1970:

- tau — receipt-gated multi-agent harness: agent work compiles to typed DAG contracts with state tracking; every handoff needs a schema-valid receipt or validator result before the next step runs. No receipt, no action. (pdf_oxide, agent-skills, and scillm are detailed under Public Work.)
- Private, regulated side: an ArangoDB agent-memory platform — approximately 219K evidence-grounded question–reasoning–answer records across 7K+ security-control records (NIST 800-53/171, CWE, MITRE ATT&CK, D3FEND, SPARTA), hybrid BM25 + vector + graph recall over 2.2M source chunks; a corpus of ~94K Lean 4 theorem statements indexed for requirements-to-proof retrieval.

Presented Agentic Cybersecurity research to Air Force Research Laboratory and received an AFRL "Hacker" challenge coin following the presentation.

15+ years of hand-coding underneath it all — today I work primarily through agentic coding, driving many harnesses and models, including my own (tau). I build the tools that make agentic development trustworthy, and I use them daily. Creative systems thinking — from leading 80+ person interactive productions — is why my AI systems are architected, not stapled together.

Buffalo, NY based (hybrid/onsite there, or remote). Open to contract engagements and full-time Principal/Staff AI Engineer, AI Architect, Machine Learning Engineer, LLM Platform, and Security/Compliance AI roles.

EXPERIENCE

Founder & Principal AI Engineer / Architect | grahamaco | Buffalo–Niagara Falls Area · Remote

Feb 2025 - Present Independent AI engineering practice. Clients are primarily export-controlled (ITAR) defense/aerospace — names withheld; publicly releasable engineering is at github.com/grahama1970.

- Built tau, a receipt-gated multi-agent orchestration harness: goals compile to typed DAG contracts; every agent handoff must produce a schema-valid receipt, hashed evidence artifact, or validator result — no receipt, no action.
- Develop a heavily diverged fork of pdf_oxide (origin: yfedoseev/pdf_oxide, MIT/Apache-2.0; independent since Mar 2026): 430 commits, ~137K lines added — Rust-core changes, full Python pipeline/plugin system, PDF-cloning fixture generation, extraction calibration, and NIST document-validation tooling.
- Authored agent-skills: 340+ reusable agent capabilities (compliance evidence mapping, document AI, LLM evaluation, retrieval) and 90+ bounded worker roles, ~85% of them gated by deterministic sanity checks — public repo, private runtime.
- Built an ArangoDB agent-memory platform (private, regulated): ~219K evidence-grounded QRA records across 7K+ security-control records; hybrid BM25 + vector + multi-hop graph recall over 2.2M source chunks; ~94K Lean 4 theorem statements indexed for requirements-to-proof retrieval.
- Core contributor to pi-mono (open-source agent framework/TUI): extension system, agent runtime, and TUI harness work, Feb–Jul 2026.

Lead Research Scientist (Agentic Formal Methods) | Self-Employed / Various Clients | Buffalo, NY

Jan 2024 - Feb 2025

- Designed verifiable agentic systems: a process-driven autoformalization workflow translating engineering requirements into Lean 4 proofs for safety-critical workflows.
- Built probabilistic-deterministic self-correcting loops where LLM generation is validated by compiler/prover feedback.
- Consulted in aerospace/defense settings under export-controlled / ITAR constraints.
- Presented Agentic Cybersecurity work to Air Force Research Laboratory; received an AFRL "Hacker" challenge coin.

Principal Data Scientist & ACERT Technical Lead | CS Group (DARPA ARCOS)

Sep 2020 - Dec 2023

- Responsible technical/project lead for CS Group's contribution to the 4-year DARPA ARCOS program (Automated Rapid Certification of Software), working alongside Honeywell, Lockheed Martin, MIT, GE Research, SRI, and other program collaborators.
- Led design and delivery of ACERT (Automated Certification of Requirements Tool): architected the ArangoDB knowledge-graph schema and LLM-assisted reasoning pipeline for multi-hop compliance verification of mission-critical software against complex certification standards.
- Worked with government and aerospace program stakeholders; split time roughly 50/50 between executive/technical briefings and hands-on architecture and code.

Founder & Principal Consultant | grahamaco / Various Clients | Buffalo, NY · Remote

2016 - 2020

- Independent data science and engineering consulting for various clients through grahamaco — the same practice resumed after the DARPA ARCOS program.

Executive Producer / Marketing Executive / Professional Composer | Los Angeles, CA

Mar 2005 - 2016

- Executive Producer, God of War: Ascension interactive campaign (Real Pie, for Sony; Webby-recognized); led cross-functional teams (80+) on large-scale interactive productions.
- Commercial composer for major consumer brands (Adidas, X-Games, Pepsi, Taco Bell, Nestle).

PUBLIC WORK (non-ITAR) — github.com/grahama1970

Client work is mostly export-controlled, so here is the public, verifiable side — including the fun stuff:

- agent-skills — my living resume: 340+ reusable agent skills, 90+ worker roles, ~85% with sanity gates. Public repo, private runtime.
- tau — receipt-gated multi-agent harness. "Agents hallucinate. Tau contains them."
- pdf_oxide — heavily diverged fork of yfedoseev's Rust PDF toolkit (430 commits, ~137K lines added: Rust-core changes, Python pipeline, PDF cloning, NIST validation).
- scillm — LLM gateway/proxy: provider routing and fallback across hosted and local models, batch pools, structured-output repair, and streaming transport for agent runtimes.
- extractor, anvil, fetcher, chatterbox voice-agent fork — supporting cast, all public.

EDUCATION

- Metis Data Science Program | 2017
- Trinity University | BS, Finance and Marketing | 1994

CORE COMPETENCIES

- Evals & Quality: LLM Evaluation, Agentic Evaluation Harnesses, Adversarial/Blind Testing, Regression Gates, Ground-Truth Fixtures, Judge Panels
- Observability & LLMOps: AI Observability, Drift Detection, Health Monitoring, Audit Trails & Receipts, Scheduled Pipelines, LLMOps
- Agentic Orchestration: Multi-Agent Systems, AI Agents, DAG Contracts, State Tracking, Tool Calling, Model Context Protocol (MCP), Bounded Agent Roles, Prompt Engineering
- LLM Platform: Large Language Models (LLM), Generative AI, LLM Gateway/Routing, Provider Fallback, Structured Outputs, Batch Inference, Guardrails
- Retrieval & Knowledge: Retrieval-Augmented Generation (RAG), GraphRAG, Knowledge Graphs, ArangoDB, Vector Databases, Hybrid BM25 + Vector Search
- Verification & Compliance: Formal Verification, Lean 4, NIST 800-53/171, MITRE ATT&CK, AI Governance
- Document AI: PDF Extraction, Layout & Table Extraction, Fixture Generation, Extraction Calibration
- ML & Platform: Machine Learning, Model Fine-Tuning, Classifier Training, Python, Rust, Docker, Linux